Homological Algebra in the Category of Modules

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Abstract

These notes present an introduction to the basic elements of homological algebra in the category of modules. They introduce chain complexes, morphisms of complexes, chain homotopy, long exact sequences in homology, projective and injective resolutions, and the construction of derived functors. As natural applications, the functors Tor and Ext are described, together with the independence of the chosen resolutions.

1 Introduction

We give an introduction to homological algebra in the category of modules, defining the concepts of chain complexes, morphisms between complexes, projective and injective resolutions, and studying their basic properties. Later we define the *n-th* left and right derived functors of an additive functor $T : {}_RM \to {}_RM$ (Ab, M_S) and, as immediate consequences, we obtain the functors Ext^n_R and Tor^R_n. Many details of the proofs will be omitted, since in many cases they are relatively simple and writing them out completely would be exhaustive.

2 Chain Complexes

Definition 2.1. A *chain complex* of modules A is a family of modules $\{A_i | i \in \mathbb{Z}, A_i \in {}_RM\}$ and a family of homomorphisms between them $\{d_{i+1} : A_{i+1} \to A_i | i \in \mathbb{Z}\}$ such that $d_i \circ d_{i+1} = 0$, or equivalently $imd_{i+1} \subset nucd_i$ for all $i \in \mathbb{Z}$. It is written as

$$A\colon\ ... \to A_{i+1} \xrightarrow{d_{i+1}} A_i \xrightarrow{d_i} A_{i-1} \to ...$$

Definition 2.2. A *complex* A is *positive* if $A_i = 0$ for all negative indices, and it is written

$$A\colon\ ... \to A_{i+1} \xrightarrow{d_{i+1}} A_i \xrightarrow{d_i} A_{i-1} \to ... \to A_1 \xrightarrow{d_1} A_0 \xrightarrow{d_0} 0.$$

Definition 2.3. A *complex* A is *negative* if $A_i = 0$ for all $i \geq 0$. Rewriting $A^n = A_{-n}$ and $d^n = d_{-n}$ for all $i \geq 0$, we obtain

$$A\colon\ 0 \xrightarrow{0} A^0 \xrightarrow{d^0} A^1 \xrightarrow{d^1} A^2 \to ... \to A^i \xrightarrow{d^i} A^{i+1} \to ...$$

with $d_{i-1} \circ d_i = 0$.

Definition 2.4. Let A be a complex of modules. The *n-th* homology module of A is $H_n(A) := nucd_n/imd_{n+1}$.

Definition 2.5. Let A and B be chain complexes of modules. A chain morphism $\varphi. : A \to B$ is a family of homomorphisms $\{\varphi : A_n \to B_n | n \in \mathbb{Z}\}$ such that the squares commute:

$$\begin{tikzcd}
\dots \arrow[r] & A_{i+1} \arrow[r, "d_{i+1}"] \arrow[d, "f_{i+1}"'] & A_i \arrow[r, "d_i"] \arrow[d, "f_i"'] & A_{i-1} \arrow[r, "d_{i-1}"] \arrow[d, "f_{i-1}"'] & \dots \\
\dots \arrow[r] & B_{i+1} \arrow[r, "d'_{i+1}"] & B_i \arrow[r, "d'_i"] & B_{i-1} \arrow[r, "d'_{i-1}"] & \dots
\end{tikzcd}$$

Let A, B, C be chain complexes and let $f. : A \to B$, $g. : B \to C$ be chain morphisms. Define $g.f. := \{g_n f_n : A_n \to B_n | n \in \mathbb{Z}\} : A \to C$; it is clearly a chain morphism since the diagrams commute.

$$\begin{tikzcd}
\dots \arrow[r] & A_{i+1} \arrow[r, "d_{i+1}"] \arrow[d, "f_{i+1}"'] & A_i \arrow[r, "d_i"] \arrow[d, "f_i"'] & A_{i-1} \arrow[r, "d_{i-1}"] \arrow[d, "f_{i-1}"'] & \dots \\
\dots \arrow[r] & B_{i+1} \arrow[r, "d'_{i+1}"] \arrow[d, "g_{i+1}"'] & B_i \arrow[r, "d'_i"] \arrow[d, "g_i"'] & B_{i-1} \arrow[r, "d'_{i-1}"] \arrow[d, "g_{i-1}"'] & \dots \\
\dots \arrow[r] & C_{i+1} \arrow[r, "d''_{i+1}"] & C_i \arrow[r, "d''_i"] & C_{i-1} \arrow[r, "d''_{i-1}"] & \dots
\end{tikzcd}$$

Proposition 2.6. *The chain complexes with the operation above form a category denoted by $Comp({}_RM)$.*

Proof Let A be a chain complex. The family $1._A := \{1_{A_n} : A_n \to A_n | n \in \mathbb{Z}\} : A \to A$ is a chain morphism. If $f. = \{f_n\} : A \to B$, then $f.1._A = \{f_n 1_{A_n} = f_n\} : A \to B$ and $f.1_A. = f.$. Analogously, if $1._B = \{1_{B_n}\} : B \to B$, then for any chain morphism $g. = \{g_n\} : B \to C$ one has $g.1_B. = g.$. If $A \xrightarrow{f.} B \xrightarrow{g.} C \xrightarrow{h.} D$ are chain morphisms, then clearly $h.(g.f). = (h.g).f.$ by associativity of the functions f_n, g_n, h_n for all $n \in \mathbb{Z}$. Therefore $Comp({}_RM)$ is a category.

If $f. = \{f_n\} : A \to B$ and $g. = \{g_n\} : A \to B$ are morphisms of chain complexes, define $f. + g.$ as $\{f_n + g_n\} : A \to B$. This is a morphism of chain complexes because

$$\delta'_{n+1}(f_{n+1} + g_{n+1}) = \delta'_{n+1} f_{n+1} + \delta'_{n+1} g_{n+1} = f_n \delta_{n+1} + g_n \delta_{n+1} = (f_n + g_n)\delta_{n+1}.$$

It follows that $f. + g. = g. + f.$. It is easy to see that if $B \xrightarrow{h_1.} C$ and $D \xrightarrow{h_2.} A$ are chain complexes, then

$$h_1.(f. + g.)h_2. = h_1.f.h_2. + h_1.g.h_2..$$

Now let $A \in Comp({}_RM)$ and define $O.$ by $O. = \{0 : 0 \to A_n | n \in \mathbb{Z}\}$, where $0 : 0 \to A_n$ is the zero morphism in ${}_RM$. Clearly $O. \in Comp({}_RM)$ and, for any morphism $f.$, one has $f. + O. = f.$.

Let $\{A^i | i \in I\} \subset Comp({}_RM)$ and define

$$\bigoplus_{i \in I} A^i$$

by

$$\textstyle\bigoplus_{i\in I} A^i : \ldots \to \bigoplus_{i\in I} A^i_{j+1} \xrightarrow{\bigoplus_{i\in I} d^i_{j+1}} \bigoplus_{i\in I} A^i_j \xrightarrow{\bigoplus_{i\in I} d^i_j} \bigoplus_{i\in I} A^i_{j-1} \to \ldots$$

where

$$\bigoplus_{i\in I} d^i_{j+1}(\{a^i_{j+1}\}_{i\in I}) = \bigoplus_{i\in I} d^i_j(\{d^i_{j+1}(a^i_{j+1})\}_{i\in I})$$
$$= \{d^i_j \circ d^i_{j+1}(a^i_{j+1})\}_{i\in I} = \{0\}_{i\in I} = 0.$$

Thus $\bigoplus_{i\in I} A^i$ is a chain complex.

Now suppose that $B \in Comp(_RM)$ and that there are chain morphisms $A^i \xrightarrow{g^i.} B$. Since $\bigoplus_{i\in I} A^i_j$ is the direct sum of the modules A^i_j, $i \in I$, and if B_i, $i \in I$, are the modules appearing in the complex B, then there exists a unique h_j for every $j \in I$ such that the following diagram commutes:

$$\begin{tikzcd} A^i_j \arrow[r, "g^i_j"] \arrow[d, "i^i_j"'] & B_j \\ \bigoplus_{i\in I} A^i_j \arrow[ur, "h_j"'] & \end{tikzcd}$$

In this way the functions h_i induce a unique chain morphism $h. = \{h_i\} : \bigoplus_{i\in I} A^i \to B$ such that $l^i.g^i. = h.$.

$$\begin{tikzcd} A^i \arrow[r, "g^i."] \arrow[d, "i^i."'] & B \\ \bigoplus_{i\in I} A^i \arrow[ur, "h."'] & \end{tikzcd}$$

Therefore $\bigoplus_{i\in I} A^i$ is a coproduct in the category $Comp(_RM)$. The product of the A^i is obtained in the same way.

Definition 2.7. Let (C, d) be a complex. A subcomplex of chains of C is a complex (A, δ) where $A_n \leq C_n$ and $\delta_n = d_n|_{A_n}$.

Definition 2.8. Let $f. : (A, d) \to (B, d')$ be a morphism of complexes. The *kernel of* $f.$, denoted by $nucf.$, is the subcomplex of A

$$nucf.\colon\ ... \to nucf_{i+1} \xrightarrow{\delta_{i+1}} nucf_i \xrightarrow{\delta_i} nucf_{i-1} \to ...$$

where $\delta_n = d_n|_{nucf_n}$.

In the same way one defines the *image of* f, denoted by $imf.$:

$$imf.\colon\ ... \to imf_{i+1} \xrightarrow{\delta'_{i+1}} imf_i \xrightarrow{\delta\,'}_i imf_{i-1} \to ...$$

where $\delta'_n = d'_n|_{imf_n}$.

Let (C, d) be a complex and (A, δ) a subcomplex. Define the quotient complex, denoted by C/A, by

$$C/A\colon\ ... \to C_{n+1}/A_{i+1} \xrightarrow{\Delta_{i+1}} C_i/A_i \xrightarrow{\Delta_i} C_{i-1}/A_{n-1} \to ...$$

where

$$\Delta_n : C_n/A_n \to C_{n-1}/A_{n-1}$$

is defined by

$$c_n + A_n \mapsto d_n(c_n) + A_{n-1}.$$

It is easy to see that Δ_n is well defined and that C/A is indeed a chain complex. Thus, if $f. : A \to B$ is a chain complex, the *cokernel of* $f.$, denoted by $conucf.$, is $B/imf.$.

Proposition 2.9. *The $nucf.$ and $conucf.$ are indeed a kernel and cokernel in the category $Com({}_RM)$. This says that $Comp({}_RM)$ is an abelian category.*

Proposition 2.10. *Let $f. : A \to B$ be a chain complex morphism. Then, for every $n \in \mathbb{Z}$, $f.$ induces a homomorphism $H_n(f.) : H_n(A) \to H_n(B)$ defined by*

$$x + imd_{n+1} \mapsto f_n(x) + imd'_{n+1}.$$

Sketch of the proof It is easy to see that $H_n(f.)$ is well defined. If $f., g. : A \to B$ are chain morphisms, then

$$H_n(f. + g.)(x + imd_{n+1}) = (f_n + g_n)(x) + imd'_{n+1} = f_n(x) + g_n(x) + imd'_{n+1}$$

$$= (f_n(x) + imd'_{n+1}) + (g_n(x) + imd'_{n+1}) = H_n(f.)(x + imd_{n+1}) + H_n(g.)(x + imd_{n+1}).$$

Thus $H_n(f. + g.) = H_n(f.) + H_n(g.)$.

Analogously, if $f. : A \to B$ and $g. : B \to C$ are chain morphisms, by writing things out one obtains $H_n(g.f.) = H_n(g.)H_n(f.)$ and $H_n(1_A.) = 1_{H_n(A)}$. Hence:

Proposition 2.11. *H_n is an additive covariant functor from $Comp({}_RM)$ to ${}_RM$ for every $n \in \mathbb{Z}$.*

Definition 2.12. Let $\varphi., \psi. : A \to B$ be chain morphisms. The morphism $\varphi.$ is homotopic to $\psi.$, denoted $\varphi. \simeq \psi.$, if there exist $h_n : A_n \to B_{n+1}$ such that $\varphi_n - \psi_n = d'_{n+1}h_n + h_{n-1}d_n$.

$$\begin{tikzcd}
{} \arrow[r] & A_{i+1} \arrow[r, "d_{i+1}"] \arrow[d, "\varphi_{i+1}-\psi_{i+1}"'] & A_i \arrow[r, "d_i"] \arrow[dl, "h_i"'] \arrow[d, "\varphi_i-\psi_i"'] & A_{i-1} \arrow[r, "d_{i-1}"] \arrow[dl, "h_{i-1}"'] \arrow[d, "\varphi_{i-1}-\psi_{i-1}"] & {} \\
{} \arrow[r] & B_{i+1} \arrow[r, "d'_{i+1}"] & B_i \arrow[r, "d'_i"] & B_{i-1} \arrow[r, "d'_{i-1}"] & {}
\end{tikzcd}$$

Proposition 2.13. *Being homotopic is an equivalence relation.*

Proposition 2.14. *If $f.$ and $g.$ are homotopic chain morphisms, then $H_n(f.) = H_n(g.)$.*

Sketch of the proof Both propositions are clear from the definition.

Let T be a covariant (contravariant) additive functor from ${}_RM$ to ${}_RM$ (Ab or M_R). If $A \in Comp({}_RM)$, let TA be

$$TA\colon \; ... \to TA_{i+1} \xrightarrow{Td_{i+1}} TA_i \xrightarrow{Td_i} TA_{i-1} \to ...$$

Then TA is a complex, since $Td_nTd_{n+1} = T0$, and $T0$ is 0 because T is additive. The contravariant additive case is analogous. If $f. : A \to B$ is a chain morphism, one defines similarly $Tf. = \{Tf_n\} : TA \to TB$, and this is also a chain morphism.

Proposition 2.15. *Let $f., g. : A \to B$ be chain morphisms with $f. \simeq g.$. If T is an additive covariant (contravariant) functor, then $Tf. \simeq Tg.$.*

Proof Since $f. \simeq g.$, we have $f_n - g_n = h_n d_{n+1} + h_{n-1} d_n$. Applying the functor gives

$$T(f_n) - T(g_n) = T(h_n)T(d_{n+1}) + T(h_{n-1})T(d_n),$$

where $T(h_n) : TA_n \to TB_{n+1}$. This is exactly the definition of $Tf. \simeq Tg.$.

The following proposition is clear by applying the preceding theorem and Proposition 2.6.

Proposition 2.16. *Let $f., g. : A \to B$ be chain morphisms with $f. \simeq g.$. If T is an additive covariant (contravariant) functor, then $H_n(Tf.) = H_n(Tg.)$.*

Definition 2.17. Let $X, X', X^{''} \in Comp(_RM)$ and let $f. : X \to X'$, $g. : X' \to X^{''}$ be chain morphisms. The sequence $O \to X \xrightarrow{f.} X' \xrightarrow{g.} X^{''} \to O$ is exact if $O \to X_{n+1} \xrightarrow{f_n} X'_n \xrightarrow{g_n} X^{''}_{n-1} \to O$ is exact for every $n \in \mathbb{Z}$.

Proposition 2.18. *If $O \to X \xrightarrow{f.} X' \xrightarrow{g.} X^{''} \to O$ is an exact sequence of complexes, then there exists an exact sequence of the form*

$$... \to H_{i+1}(X) \xrightarrow{H_{i+1}(g)} H_{i+1}(X') \xrightarrow{\Delta_{i+1}} H_i(X') \xrightarrow{H_i(f)} H_i(X) \xrightarrow{H_i(g)} H_i(X^{''}) \to ...$$

Sketch of the proof Consider the following diagram.

$$\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d] & \vdots \arrow[d] & \\
0 \arrow[r] & X'_{n+1} \arrow[r, "f_{n+1}"] \arrow[d, "d'_{n+1}"] & X_{n+1} \arrow[r, "g_{n+1}"] \arrow[d, "d_{n+1}"] & X''_{n+1} \arrow[r] \arrow[d, "d''_{n+1}"] & 0 \\
0 \arrow[r] & X'_n \arrow[r, "f_n"] \arrow[d, "d'_n"] & X_n \arrow[r, "g_n"] \arrow[d, "d_n"] & X''_n \arrow[r] \arrow[d, "d''_n"] & 0 \\
0 \arrow[r] & X'_{n-1} \arrow[r, "f_{n-1}"] \arrow[d, "d'_{n-1}"] & X_{n-1} \arrow[r, "g_{n-1}"] \arrow[d, "d_{n-1}"] & X''_{n-1} \arrow[r] \arrow[d, "d''_{n-1}"] & 0 \\
 & \vdots & \vdots & \vdots &
\end{tikzcd}$$

Let $c_n^{''} \in nucd_n^{''}$. Since g_n is an epimorphism, there exists $c_n \in X_n$ such that $g_n(c_n) = c_n^{''}$. Then $d_n^{''} g_n(c_n) = 0 = g_{n-1} d_n(c_n)$, because the squares commute, and therefore $d_n(c_n) \in nucg_{n-1} = imf_{n-1}$. Since f_{n-1} is injective, there exists a unique $c'_{n-1} \in X'_{n-1}$ such that $f_{n-1}(c'_{n-1}) = d_n(c_n)$. By commutativity of the squares, it is easy to see that $c'_{n-1} \in nucd'_{n-1}$. Define

$$\Delta_n : H_n(X^{''}) \to H_n(X')$$

by

$$c_n + imd_{n+1}^{''} \mapsto c'_{n-1} + imd'_n,$$

where $f_{n-1}(c'_{n-1}) = d_n(c_n)$.

This relation does not depend on the choice of c_n such that $g_n(c_n) = c''_n$. Indeed, if $c_n, e_n \in X_n$ satisfy $g_n(c_n) = g_n(e_n) = c''_n$, then $g_n(c_n - e_n) = 0$, so $(c_n - e_n) \in nucg_n = imf_n$. Thus $c_n - e_n = f_n(x'_n)$, and applying d_n gives

$$d_n(c_n - e_n) = d_n f_n(x'_n) = f_{n-1} d'_n(x'_n).$$

Hence $d_n(c_n - e_n) \in imf_{n-1}$. Since also $d_n(c_n) = f_{n-1}(c'_{n-1})$ and $d_n(e_n) = f_{n-1}(e'_{n-1})$, we obtain

$$f_{n-1}^{-1} d_n(c_n - e_n) = f_{n-1}^{-1} d_n(c_n) - f_{n-1}^{-1} d_n(e_n) \in imd'_n.$$

Therefore

$$f_{n-1}^{-1} d_n(c_n) + imd'_n = f_{n-1}^{-1} d_n(e_n) + imd'_n,$$

so the definition is independent of the choice of c_n.

It is well defined. If

$$c''_n + imd''_{n+1} = d''_n + imd''_{n+1},$$

then

$$c''_n - d''_n = d''_{n+1} g_{n+1}(x''_{n+1}),$$

and hence

$$c''_n - d''_n = d''_{n+1} g_{n+1}(x''_{n+1}) = g_n d_{n+1}(x_{n+1}) = g_n(c_n - e_n).$$

By the preceding part,

$$f_{n-1}^{-1} d_n d_{n+1}(x_{n+1}) + imd'_n = f_{n-1}^{-1} d_n(c_n - e_n) + imd'_n.$$

This says that

$$0 + imd'_n = f_{n-1}^{-1} d_n(c_n) - f_{n-1}^{-1} d_n(e_n) + imd'_n,$$

and finally

$$f_{n-1}^{-1} d_n(c_n) + imd'_n = f_{n-1}^{-1} d_n(e_n) + imd'_n.$$

Thus Δ_n is well defined. Checking that it is a module morphism and that the sequence is exact is routine.

3 Projective Resolutions

Definition 3.1. Let $A \in {}_R M$. An exact sequence of the form

$$P : ... \to P_{i+1} \xrightarrow{d_{i+1}} P_i \xrightarrow{d_i} P_{i-1} \to ... \to P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} A \to 0$$

in which every module P_n is projective is called a projective resolution of A.

The sequence, no longer required to be exact, of the form

$$P_A : ... \to P_{i+1} \xrightarrow{d_{i+1}} P_i \xrightarrow{d_i} P_{i-1} \to ... \to P_1 \xrightarrow{d_1} P_0 \to 0$$

is called the reduced complex of the projective resolution P.

Proposition 3.2. *Every module A has a projective resolution.*

Proof Since every module A is the homomorphic image of some projective module P_0, we have

$$P_0 \to A \to 0.$$

Suppose inductively that we have found an exact sequence of the form

$$...P_n \xrightarrow{d_n} P_{n-1} \to ... \to P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} A \to 0.$$

Let $K_n = nucd_n$ and consider $K_n \xrightarrow{i_n} P_n \xrightarrow{d_n} P_{n-1}$ with the inclusion. Using again that every module is a homomorphic image of a projective module, we have

$$P_{n+1} \xrightarrow{d'_{n+1}} K_n \xrightarrow{i_n} P_n \xrightarrow{d_n} P_{n-1}.$$

Let $d_{n+1} = i_n d'_{n+1}$. It is easy to see that

$$P_{n+1} \xrightarrow{d_{n+1}} P_n \xrightarrow{d_n} P_{n-1} \to ... \to P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} A \to 0$$

is exact. Therefore every module A has a projective resolution.

Definition 3.3. Let P, P' be chain complexes, let $A, B \in {}_RM$, and let $f : A \to B$ be a module homomorphism, where P and P' are not necessarily projective resolutions of A and B, respectively. A chain map over the homomorphism f is a morphism between X_A and X'_B, $f. = \{f_n\}_{n\geq 0} : X_A \to X'_B$, such that $d_0 f = d'_0 f_0$.

$$\begin{tikzcd}
{} \arrow[r] & P_{i+1} \arrow[r, "d_{i+1}"] & P_i \arrow[r, "d_i"] & \dots \arrow[r] & P_1 \arrow[r, "d_1"] & P_0 \arrow[r, "d_0"] & A \arrow[r] \arrow[d, "f"] & 0 \\
{} \arrow[r] & P'_{i+1} \arrow[r, "d'_{i+1}"] & P'_i \arrow[r, "d'_i"] & \dots \arrow[r] & P'_1 \arrow[r, "d'_1"] & P'_0 \arrow[r, "d'_0"] & B \arrow[r] & 0
\end{tikzcd}$$

Proposition 3.4. *Let $A, B \in {}_RM$, let $f : A \to B$ be a module homomorphism, and let P and P' be projective resolutions of A and B, respectively. Then there exists a chain map over f, and any two such maps are homotopic.*

Proof Consider the following diagram:

$$\begin{tikzcd}
{} \arrow[r] & P_{i+1} \arrow[r, "d_{i+1}"] & P_i \arrow[r, "d_i"] & \dots \arrow[r] & P_1 \arrow[r, "d_1"] & P_0 \arrow[r, "d_0"] & A \arrow[r] \arrow[d, "f"] & 0 \\
{} \arrow[r] & P'_{i+1} \arrow[r, "d'_{i+1}"] & P'_i \arrow[r, "d'_i"] & \dots \arrow[r] & P'_1 \arrow[r, "d'_1"] & P'_0 \arrow[r, "d'_0"] & B \arrow[r] & 0
\end{tikzcd}$$

Since P_0 is projective, there exists $f_0 : P_0 \to P'_0$ such that $d'_0 f_0 = f d_0$.

Now suppose there exist $f_j : P_j \to P'_j$ such that $d'_j f_j = f_{j-1} d_j$ for all $1 \leq j \leq i$. Consider the homomorphism $d_{i+1} f_i : P_{i+1} \to P'_i$. Since $d'_i(d_{i+1} f_i) = (d'_i d_{i+1}) f_i$ and, by commutativity of the squares,

$$(d'_i d_{i+1}) f_i = (d_{i+1} d_i) f_{i-1} = 0,$$

we have

$$\begin{tikzcd}
& P_{i+1} \arrow[dl, dashed] \arrow[d, "d_{i+1} f_i"] & \\
P'_{i+1} \arrow[r, "d'_{i+1}"'] & P'_i \arrow[r, "d'_i"'] & P'_{i-1}
\end{tikzcd}$$

and, since P_{i+1} is projective, there exists $f_{i+1} : P_{i+1} \to P'_{i+1}$ such that

$$d_{i+1} f_i = d'_{i+1} f_{i+1}.$$

Therefore there exists a chain map over f.

Let $f., g. : P_A \to P'_B$ be chain maps over f. Since $d'_0 f_0 = f d_0 = d'_0 g_0$, we have $d'_0(f_0 - g_0) = 0$. Hence there exists $h_0 : P_0 \to P'_1$ such that

$$f_0 - g_0 = d'_1 h_0.$$

Let $h_{-1} = 0$. Suppose that $f_j - g_j = d'_{j+1} h_j + h_{j-1} d_j$ for $0 \leq j \leq i$, and consider

$$f_{i+1} - g_{i+1} - h_i d_{i+1} : P_{i+1} \to P'_{i+1}.$$

$$\begin{tikzcd}
 & P_{i+1} \arrow[dl, dashed] \arrow[d, "f_{i+1} - g_{i+1} - h_i d_{i+1}"] & \\
P'_{i+2} \arrow[r, "d'_{i+2}"'] & P'_{i+1} \arrow[r, "d'_{i+1}"'] & P'_i
\end{tikzcd}$$

Applying d_{i+1} gives

$$d'_{i+1}(f_{i+1} - g_{i+1} - h_i d_{i+1}) = d'_{i+1}(f_{i+1}) - d'_{i+1}(g_{i+1}) - d'_{i+1} h_i d_{i+1}$$

$$= f_i d_{i+1} - g_i d_{i+1} - d'_{i+1} h_i d_{i+1} = (f_i - g_i - d'_{i+1} h_i) d_{i+1} = (h_{i-1} d_i) d_{i+1} = 0.$$

Thus there exists a homomorphism $h_{i+1} : P_{i+1} \to P'_{i+2}$ such that

$$f_{i+1} - g_{i+1} - h_i d_{i+1} = d'_{n+2} h_{n+1}.$$

This says that $f. \simeq g.$

Recall that if P and P' are two projective resolutions of A, then there exists a chain map $f. : P_A \to P'_A$ over 1_A and a chain map $g. : P'_A \to P_A$ over 1_A. By the preceding theorem, $g.f. : P_A \to P_A$ and $f.g. : P'_A \to P'_A$ are homotopic to 1_{P_A} and $1_{P'_A}$, respectively.

4 Injective Resolutions

Definition 4.1. Let $A \in {}_R M$. An exact sequence of the form

$$E : 0 \to A \xrightarrow{d^{-1}} E^0 \xrightarrow{d^0} E^1 \xrightarrow{d^1} E^2 \to ... \to E^i \xrightarrow{d^i} E^{i+1} \to ...$$

in which every module E^n is injective is called an injective resolution of A.

The sequence, no longer required to be exact, of the form

$$E_A : 0 \to E^0 \xrightarrow{d^0} E^1 \xrightarrow{d^1} E^2 \to ... \to E^i \xrightarrow{d^i} E^{i+1} \to ...$$

is called the reduced complex of the injective resolution E.

Analogues of the theorems on projective resolutions are the following.

Proposition 4.2. *Every module A has an injective resolution.*

Definition 4.3. Consider the following diagram:

$$\begin{array}{ccccccccccccc}
0 & \longrightarrow & A & \xrightarrow{d^{-1}} & E^0 & \xrightarrow{d^0} & E^1 & \longrightarrow & \dots & \longrightarrow & E^i & \longrightarrow & \dots \\
 & & \downarrow{\scriptstyle f} & & & & & & & & & & \\
0 & \longrightarrow & B & \xrightarrow{d_{\prime}^{-1}} & E_{\prime}^0 & \xrightarrow{d_{\prime}^0} & E_{\prime}^1 & \longrightarrow & \dots & \longrightarrow & E_{\prime}^i & \longrightarrow & \dots
\end{array}$$

where $E, E_{\prime}$ are chain complexes, $A, B \in {}_RM$, and $f : A \to B$ is a module homomorphism, with E and $E_{\prime}$ not necessarily injective resolutions of A and B, respectively. A chain map over the homomorphism f is a morphism between E_A and $E_{\prime B}$, $f. = \{f^n\}_{n\geq 0} : E_A \to E_{\prime B}$, such that $d^{-1}f^0 = d_1^{-1}f$.

Proposition 4.4. *Let $A, B \in {}_RM$, let $f : A \to B$ be a module homomorphism, and let E and E_1 be injective resolutions of A and B, respectively. Then there exists a chain map over f, and any two such maps are homotopic.*

5 Derived Functors

In this section $T : {}_RM \to {}_RM$ (Ab, M_S) is a covariant additive functor and $T' : {}_RM \to {}_RM$ (Ab, M_S) is a contravariant additive functor.

Let $A, B, C \in {}_RM$, let $f : A \to B$ and $g : B \to C$ be module homomorphisms, and let P, P', P'' be projective resolutions of A, B, C, respectively. Consider the following diagram:

$$\begin{array}{cccccccccccccc}
\longrightarrow & P_{i+1} & \xrightarrow{d_{i+1}} & P_i & \xrightarrow{d_i} & \dots & \longrightarrow & P_1 & \xrightarrow{d_1} & P_0 & \xrightarrow{d_0} & A & \longrightarrow & 0 \\
 & & & & & & & & & & & \downarrow{\scriptstyle f} & & \\
\longrightarrow & P'_{i+1} & \xrightarrow{d'_{i+1}} & P'_i & \xrightarrow{d'_i} & \dots & \longrightarrow & P'_1 & \xrightarrow{d'_1} & P'_0 & \xrightarrow{d'_0} & B & \longrightarrow & 0 \\
 & & & & & & & & & & & \downarrow{\scriptstyle g} & & \\
\longrightarrow & P''_{i+1} & \xrightarrow{d''_{i+1}} & P''_i & \xrightarrow{d''_i} & \dots & \longrightarrow & P''_1 & \xrightarrow{d''_1} & P''_0 & \xrightarrow{d''_0} & C & \longrightarrow & 0
\end{array}$$

Define $L_nT(A) = H_n(TP_A) = nucTd_n/imTd_{n+1}$. By Theorem 3.2 there exists a chain map $f. = \{f_n\}_{n\geq 0} : P_A \to P'_B$ over f. Then $Tf. = \{Tf_n\}_{n\geq 0} : TP_A \to TP'_B$ is a chain map over Tf, and we set

$$L_nT(f) := H_n(Tf.) : L_nT(A) \to L_nT(B),$$

defined by

$$x + imTd_{n+1} \longmapsto Tf_n(x) + imTd'_{n+1},$$

with $x \in nucTd_n$.

This definition does not depend on the morphism between the complexes P and P' over f, because if $g.$ is another chain map over f, then $f. \simeq g.$. By Proposition 2.7, $Tf. \simeq Tg.$, and by Theorem 2.8, $H_n(Tf.) = H_n(Tg.)$.

Proposition 5.1. *L_nT is an additive covariant functor from ${}_RM$ to ${}_RM$ (Ab, M_S) for every $n \geq 0$.*

Proof $1_{P_A}. = \{1_{P_n}\}_{n\geq 0} : P_A \to P_A$ is a chain morphism over 1_A. Thus

$$L_nT(1_A)(x + imTd_{n+1}) = 1_{TP_n}(x) + imTd_{n+1} = x + imTd_{n+1} = 1_{L_nT(A)}(x + imT_{n+1}).$$

Hence

$$L_nT(1_A) = 1_{L_nT(A)}.$$

Let $g : B \to C$ be a module homomorphism. By Theorem 3.2 there exists a chain morphism $g. = \{g_n\}_{n\geq 0} : P'_B \to P''_C$ over g. Then $g.f.$ is a chain morphism over gf, so

$$L_nT(gf)(x + imTd_{n+1}) = T(g_nf_n)(x) + imTd''_{n+1}$$

$$= (Tg_nTf_n)(x) + imTd''_{n+1} = L_nT(g)(Tf_n(x) + imTd'_{n+1}) = L_nT(g)L_nT(f)(x + imTd_{n+1}).$$

Therefore

$$L_nT(gf) = L_nT(g)L_nT(f).$$

Now let $h : A \to B$ and let $h. = \{h_n\}_{n\geq 0} : P_A \to P'_B$ be a chain morphism over h. Then $f. + h.$ is a chain map over $f + h$, and

$$L_nT(f + h)(x + imTd_{n+1}) = T(f_n + h_n)(x) + imTd'_{n+1}$$

$$= (Tf_n + Th_n)(x) + imTd'_{n+1} = (Tf_n(x) + imTd'_{n+1}) + (Th_n(x) + imTd'_{n+1})$$

$$= L_nT(f)(x + imTd_{n+1}) + L_nT(h)(x + imTd_{n+1}).$$

Therefore

$$L_nT(f + h) = L_nT(f) + L_nT(h).$$

Analogously define

$$R^nT'(A) = H_n(T'P_A).$$

If $f : A \to B$, by Theorem 3.2 there exists a chain map $f. = \{f^n\}_{n\geq 0} : P_A \to P'_B$ over f. Then $T'f. = \{T'f_n\}_{n\geq 0} : T'P'_B \to T'P_A$ is a chain map over $T'f$. Let

$$R^nT'(f) = H_n(T'f.) : R^nT'(B) \to R^nT'(A),$$

with correspondence rule

$$x + imT'd'_n \longmapsto T'f_n(x) + imT'd_n,$$

where $x \in nucT'd'_{n+1}$.

Analogously to the preceding proposition, we have:

Proposition 5.2. *R^nT' is an additive contravariant functor from ${}_RM$ to ${}_RM$ (Ab, M_S) for every $n \geq 0$.*

Sketch of the proof The proof is very similar to the preceding one, remembering that T' is contravariant and that the arrows are reversed.

It is natural to ask what would have happened if, instead of taking projective resolutions, one had taken injective resolutions. For this, let $A, B, C \in {}_RM$, let $f : A \to B$ and $g : B \to C$ be module homomorphisms, let $E, E_{\prime}, E_{\prime\prime}$ be injective resolutions of A, B, C, and let $E_A \xrightarrow{f.} E_{\prime B}$ and $E_{\prime B} \xrightarrow{g.} E_{\prime C}$ be chain maps over f and g, respectively:

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r, "d^{-1}"] \arrow[d, "f"] & E^0 \arrow[r, "d^0"] & E^1 \arrow[r] & \dots \arrow[r] & E^i \arrow[r] & \dots \\
0 \arrow[r] & B \arrow[r, "d_{\prime}^{-1}"] \arrow[d, "g"] & E_{\prime}^0 \arrow[r, "d_{\prime}^0"] & E_{\prime}^1 \arrow[r] & \dots \arrow[r] & E_{\prime}^i \arrow[r] & \dots \\
0 \arrow[r] & C \arrow[r, "d_{\prime\prime}^{-1}"] & E_{\prime\prime}^0 \arrow[r, "d_{\prime\prime}^0"] & E_{\prime\prime}^1 \arrow[r] & \dots \arrow[r] & E_{\prime\prime}^i \arrow[r] & \dots
\end{tikzcd}$$

Define

$$R^nT(A) := H_n(TE_A), \qquad R^nT(f) := H_n(Tf.),$$

and

$$L_nT'(A) := H_n(T'E_A), \qquad L_nT'(f) := H_n(T'f.) \quad \forall n \geq 0.$$

Then we have the following propositions.

Proposition 5.3. *R^nT is an additive covariant functor from ${}_RM$ to ${}_RM$ (Ab, M_S) for every $n \geq 0$.*

Proposition 5.4. *L_nT' is an additive contravariant functor from ${}_RM$ to ${}_RM$ (Ab, M_S) for every $n \geq 0$.*

In these definitions fixed projective (injective) resolutions of a module A were chosen. We now show that the constructions do not depend on these choices.

Proposition 5.5. *Let $P, \overline{P}$ be projective resolutions of A, and let $L_nT(A) = H_n(TP_A)$ and $\overline{L}_nT(A) = H_n(T\overline{P}_A)$. Then $L_nT(A) \cong \overline{L}_nT(A)$.*

Proof Consider the following diagram:

$$\begin{tikzcd}
{} \arrow[r] & P_{i+1} \arrow[r, "d_{i+1}"] & P_i \arrow[r, "d_i"] & \dots \arrow[r] & P_1 \arrow[r, "d_1"] & P_0 \arrow[r, "d_0"] & A \arrow[r] \arrow[d, "1_A"] & 0 \\
{} \arrow[r] & \overline{P}_{i+1} \arrow[r, "\overline{d}_{i+1}"] & \overline{P}_i \arrow[r, "\overline{d}_i"] & \dots \arrow[r] & \overline{P}_1 \arrow[r, "\overline{d}_1"] & \overline{P}_0 \arrow[r, "\overline{d}_0"] & A \arrow[r] & 0
\end{tikzcd}$$

Let $P_A \xrightarrow{f.} \overline{P}_A$ and $\overline{P}_A \xrightarrow{g.} P_A$ be chain morphisms over 1_A. Then $P_A \xrightarrow{f.g.} P_A$ and $\overline{P}_A \xrightarrow{g.f.} \overline{P}_A$ are chain morphisms over 1_A. But $\overline{P}_A \xrightarrow{1.\overline{P}_A} \overline{P}_A$ and $P_A \xrightarrow{1.P_A} P_A$ are also chain morphisms over 1_A. Therefore, by Proposition 3.2, these morphisms are respectively homotopic, so $g.f. \simeq 1_{P_A}$ and $f.g. \simeq 1_{\overline{P}_A}$. By Proposition 2.7, $T(f.g.) = Tf.Tg. \simeq T1.{}_{\overline{P}_A} = 1.{}_{T\overline{P}_A}$, and analogously $T(g.f.) = Tg.Tf. \simeq 1.{}_{TP_A}$. By Proposition 2.8,

$$H_n(Tg.Tf.) = H_n(Tf.)H_n(Tg.) = H_n(1.{}_{TP_A}) = 1_{H_n(TP_A)}.$$

In the same way,

$$H_n(Tf.)H_n(Tg.) = 1_{H_n(T\overline{P}_A)}.$$

Thus

$$\overline{L}_nT(A) \xrightarrow{H_n(Tg.)} L_nT(A) \xrightarrow{H_n(Tf.)} \overline{L}_nT(A)$$

and

$$L_nT(A) \xrightarrow{H_n(Tf.)} \overline{L}_nT(A) \xrightarrow{H_n(Tg.)} L_nT(A)$$

are the identity on $\overline{L}_nT(A)$ and $L_nT(A)$, respectively. Therefore

$$\overline{L}_nT(A) \cong L_nT(A).$$

Analogous to these propositions, we have:

Proposition 5.6. *Let $P, \overline{P}$ be projective resolutions. If $R^nT'(A) = H_n(T'P_A)$ and $\overline{R}^nT'(A) = H_n(T'\overline{P}_A)$, then $R^nT'(A) \cong \overline{R}^nT'(A)$ for every $n \geq 0$.*

Proposition 5.7. *Let $E, \overline{E}$ be injective resolutions of A. If $R^nT(A) = H_n(TE_A)$ and $\overline{R}^nT(A) = H_n(T\overline{E}_A)$, then $R^nT(A) \cong \overline{R}^nT(A)$ for every $n \geq 0$.*

Proposition 5.8. *Let $E, \overline{E}$ be injective resolutions of A. If $R^nT(A) = H_n(TE_A)$ and $\overline{R}^nT(A) = H_n(T\overline{E}_A)$, then $R^nT(A) \cong \overline{R}^nT(A)$ for every $n \geq 0$.*

Propositions 5.6, 5.7, and 5.8 are proved in a way similar to Proposition 5.5.

Proposition 5.9. *The following are naturally equivalent:*

$$L_nT \text{ and } \overline{L}_nT,\ R^nT' \text{ and } \overline{R}^nT',\ L_nT' \text{ and } \overline{L}_nT',\ R^nT \text{ and } \overline{R}^nT.$$

Proof We prove only the natural equivalence of L_nT and $\overline{L}_nT$.

Let $h : A \to B$ be a module homomorphism, let $P, \overline{P}, Q, \overline{Q}$ be projective resolutions of A and B, respectively, let $P_A \xrightarrow{h.} Q_B$ and $\overline{P}_A \xrightarrow{\overline{h}.} \overline{Q}_B$ be chain maps over h, and let $Q_B \xrightarrow{k.} \overline{Q}_B$ be a chain map over 1_B. Then $P_A \xrightarrow{k.h.} \overline{Q}_B$ and $P_A \xrightarrow{\overline{h}.f.} \overline{Q}_B$ are chain maps over h. This says that

$$k.h. \simeq \overline{h}.f.$$

Therefore

$$T(k.h.) = Tk.Th. \simeq T(\overline{h}.f.) = T\overline{h}.Tf.$$

Thus

$$H_n(Tk.Th.) = H_n(Tk.)H_n(Th.) = H_n(T\overline{h}.Tf.) = H_n(T\overline{h}.)H_n(Tf.).$$

This says that the following diagram is commutative:

$$\begin{tikzcd}
L_nT(A) \arrow[r, "H_n(Th.)"] \arrow[d, "H_n(Tf.)"] & L_nT(B) \arrow[d, "H_n(Tk.)"] \\
\overline{L}_nT(A) \arrow[r, "H_n(T\overline{h}.)"] & \overline{L}_nT(B)
\end{tikzcd}$$

Definition 5.10. L_nT and R^nT are called the n-th left and right derived functors of T, respectively.

Definition 5.11. Let $N \in M_R$ and $M \in {}_RM$. Then

$Tor_n^R(N, \_) = L_nT(\_)$ for $T = N \otimes \_$,

$tor_n^R(\_, M) = L_nT(\_)$ for $T = \_ \otimes M$,

$Ext_R^n(\_, M) = R^nT'(\_)$ for $T' = Hom(\_, M)$,

$ext_R^n(M, \_) = R^nT(\_)$ for $T = Hom(M, \_)$.

A fact that will not be proved is that if $A \in M_R$ and $B \in {}_RM$, then $Tor_n^R(A, B) \cong tor_n^R(A, B)$; and if $C, D \in {}_RM$, then $Ext_R^n(C, D) \cong ext_R^n(C, D)$.